

Few-Shot Federated Learning Framework for Rare Event Prediction in Heterogeneous IoT Sensor Networks

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Abstract: The increasing reliance on the Internet of Things (IoT) has amplified the need for accurate and timely prediction of rare events, such as equipment failures or security threats, within vast and diverse sensor networks. Traditional centralized machine learning methods are often impractical due to significant privacy concerns, high communication costs, and network latency associated with transmitting massive volumes of sensitive data. Furthermore, the inherent heterogeneity of IoT devices, which possess varying computational capabilities and data distributions, renders a singular, uniform model ineffective. This paper introduces a novel Few-Shot Federated Learning (FS-FL) framework designed to tackle these challenges. The framework leverages federated learning to enable collaborative model training among distributed IoT devices, ensuring data privacy by preventing the sharing of raw data. It incorporates few-shot learning techniques, allowing the model to effectively learn from the scarce data samples that characterize rare events. A meta-learning-based aggregation strategy is also proposed to adapt the global model to the unique data characteristics of each device, enhancing its performance across the heterogeneous network. Through extensive simulations, the FS-FL framework demonstrates superior prediction accuracy for rare events compared to existing methods, confirming its effectiveness, robustness, and scalability for real-world IoT applications.

Keywords: *Federated Learning, Few-Shot Learning, Internet of Things (IoT), Rare Event Prediction, Heterogeneous Networks, Machine Learning, Data Privacy, Anomaly Detection*

I. INTRODUCTION

1.1 Background and Motivation

The proliferation of the Internet of Things (IoT) has transformed modern industries, enabling real-time monitoring, automation, and intelligent decision-making across diverse domains such as smart cities, industrial automation, healthcare, and critical infrastructure

management. IoT networks consist of numerous interconnected sensors and devices that continuously generate vast volumes of data. While this abundance of information has the potential to drive predictive insights, it also presents significant challenges for traditional machine learning techniques, particularly when dealing with **rare events** such as equipment failures, security breaches, or critical anomalies. Rare events, by definition, occur infrequently but often have high-impact consequences. Predicting these events accurately is crucial to preventing costly downtime, mitigating risks, and ensuring the reliability of IoT-enabled systems. However, these events are inherently difficult to model because of the limited availability of labeled data, which restricts the effectiveness of conventional supervised learning approaches.

1.2 Challenges in IoT-Based Rare Event Prediction

Predicting rare events in heterogeneous IoT networks presents several interconnected challenges:

1. **Data Privacy and Security Concerns:** IoT devices frequently generate sensitive or proprietary data. Centralized machine learning methods, which rely on aggregating all data into a single server, pose significant privacy risks and are often infeasible in contexts such as healthcare, critical infrastructure, and industrial monitoring.
2. **Network Heterogeneity:** IoT networks are composed of devices with widely varying computational capabilities, storage capacities, and communication bandwidths. A singular, uniform model cannot optimally handle the differences in local data distributions and device constraints.
3. **Limited Labeled Data for Rare Events:** Rare events are characterized by sparse data points, making it challenging to train traditional machine learning models without overfitting or underperforming. Standard data-hungry

approaches fail to generalize well under such conditions.

4. **Communication and Latency Constraints:** Centralized model training requires transmitting large amounts of sensor data to a central server, which introduces latency, consumes bandwidth, and increases the likelihood of network congestion. This is particularly problematic in large-scale IoT deployments.

Rare events are difficult to predict due to extreme class imbalance, insufficient labeled samples, and non-stationary sensor behavior. Traditional models tend to bias toward dominant normal classes, leading to poor generalization and high false-negative rates for rare but critical events. Sharing raw sensor data with a central server introduces privacy risks, increases communication overhead, and creates single points of failure. It also violates regulatory constraints in sensitive domains and becomes impractical for large-scale IoT networks due to latency and bandwidth limitations.

1.3 Federated and Few-Shot Learning for IoT

Federated Learning (FL) has emerged as a promising paradigm for decentralized, privacy-preserving model training. In FL, each IoT device trains a local model using its own data, and only model parameters or updates are shared with a central server. This approach mitigates privacy risks, reduces communication overhead, and respects regulatory constraints while enabling collaborative learning across distributed networks.

However, FL alone cannot fully address the challenge of rare event prediction. Due to the scarcity of event-specific data on individual devices, models trained locally often struggle to generalize. To address this, few-shot learning (FSL) techniques can be integrated into the federated framework. FSL enables models to learn effectively from very limited labeled samples, making it ideal for capturing rare and critical events in heterogeneous IoT environments.

1.4 Proposed Approach: Few-Shot Federated Learning Framework

This paper introduces a Few-Shot Federated Learning (FS-FL) framework for rare event prediction in heterogeneous IoT sensor networks. The proposed framework combines the strengths of federated learning and few-shot learning while incorporating a meta-learning-based aggregation strategy to adapt the global model to the specific data distributions of individual devices. Key contributions include:

- **Privacy-Preserving Collaborative Learning:** FS-FL enables devices to collaboratively train a global model without sharing raw sensor data, ensuring user privacy and regulatory compliance.
- **Few-Shot Adaptation:** The framework is capable of learning from scarce labeled samples associated with rare events, overcoming data sparsity challenges.
- **Meta-Learning-Based Aggregation:** By leveraging meta-learning principles, the global model is dynamically adapted to the heterogeneity

of local data, improving overall prediction accuracy.

- **Scalability and Robustness:** Extensive simulations demonstrate the framework's capability to handle large-scale IoT networks while maintaining high performance for rare event prediction.

A single global model is insufficient for heterogeneous IoT networks because devices exhibit diverse sensing modalities, operating conditions, and data distributions. Such heterogeneity leads to model bias and degraded local performance, which is mitigated in FS-FL through meta-learning-based personalization.

1.5 Organization of the Paper

The remainder of this paper is organized as follows. Section 2 reviews related work on rare event prediction, federated learning, and few-shot learning. Section 3 presents the architecture and methodological details of the FS-FL framework, including the meta-learning-based aggregation strategy. Section 4 describes the experimental setup and evaluation metrics. Section 5 discusses simulation results, comparative analyses, and performance evaluation. Finally, Section 6 concludes the paper and outlines future research directions.

II. LITERATURE REVIEW

A. IoT Data Quality, Missingness, and Time-Series Imputation

Data quality and missing values are fundamental challenges in IoT sensor networks. Classical statistical treatments of missing data remain foundational [8], but modern IoT applications require scalable, structure-aware imputation that leverages temporal and relational patterns. Recent work has therefore focused on graph-based representations of sensor networks to exploit spatial/relational dependencies when imputing multivariate time series. Chen et al. [1] explored both explicit and implicit graph learning for multivariate time-series imputation, showing that graph-aware models can significantly improve reconstruction accuracy by modeling inter-variable relationships. Wang et al. [9] advanced this direction with position-aware graph neural networks specifically for networked time series, demonstrating the value of location/position cues in imputation quality. Complementary methods such as GRIN [2] use graph neural networks (GNNs) to model multivariate dependencies and successfully handle complex missing-data patterns. Collectively, these studies indicate that graph-structured modeling is effective for sensor data imputation and provides a useful backbone for downstream anomaly detection and prediction tasks.

B. Graph Neural Networks and Time-Series Tasks

Graph neural networks have become central to time-series modeling where relationships among variables or devices matter. Surveys [6], [7] summarize the rapid adoption of GNNs across forecasting, classification, imputation, and anomaly detection tasks—highlighting architectures that combine temporal operators (RNNs, attention, diffusion

convolutions) with graph convolutions. Li et al. [13] introduced diffusion convolutional recurrent neural networks (DCRNN) for traffic forecasting, an influential architecture that integrates diffusion graph convolutions with recurrent temporal modeling; its style of combining spatial-temporal operators is directly applicable to IoT sensor networks. More recent work [3] improves GNN-based anomaly detection with indicator attention mechanisms and Bayesian optimization, illustrating two trends: (1) adding attention or indicator modules to focus on salient temporal patterns, and (2) using automated hyperparameter/architecture search to tune GNNs for anomaly tasks. Domain applications of GNNs in infrastructure [10] and IoT data quality [4] further emphasize that graph-enabled methods can improve robustness and interpretability in networked sensing environments.

C. Anomaly and Rare-Event Detection in IoT

Detecting rare events in IoT is challenging not only because of class imbalance (few positive samples) but also due to resource constraints on edge devices and communication limits. Guo et al. [5] addressed energy-efficient anomaly detection for multivariate IoT time series, demonstrating that model and algorithmic choices can materially reduce on-device energy while preserving detection accuracy. Power- and network-efficiency concerns raised in works like Sivasankar et al. [12] reflect a broader need: anomaly detection frameworks must be communication-aware and computationally lightweight to be deployable across heterogeneous IoT nodes. These pressures motivate designs that minimize raw-data transfer (to preserve bandwidth and privacy) and reduce on-device computation without sacrificing model efficacy.

D. Federated Learning, Personalization, and Meta-Learning

Federated learning (FL) is a natural paradigm for privacy-preserving distributed model training in IoT, because it keeps raw data on-device and exchanges model updates instead. However, standard FL algorithms typically assume abundant labeled data per client and often yield suboptimal performance under severe class imbalance or non-iid local distributions—conditions typical of rare-event scenarios. Meta-learning and personalization techniques have been proposed to bridge this gap by enabling rapid client-side adaptation from limited data; although the references in this review do not directly present federated/meta-learning frameworks, the need for meta-aggregation mechanisms follows directly from the heterogeneity documented in GNN/time-series and resource-aware anomaly detection literature [3], [5], [6]. The proposed meta-learning-based aggregation in FS-FL aligns with this trend: meta-parameters serve as a transferable initialization that devices can fine-tune with very few local samples, enabling few-shot adaptation.

E. Few-Shot Learning for Scarce-Event Regimes

Few-shot learning (FSL) techniques—especially metric-based approaches such as prototypical networks—are

explicitly designed to generalize from very small labeled support sets and are therefore well-suited to rare-event prediction. While the reviewed literature does not directly propose federated few-shot frameworks for IoT, the prior works on imputation [1], [2], [9], GNNs for anomaly detection [3], and energy-aware models [5], [12] imply a clear opportunity: combining FSL with FL can enable robust rare-event detection while respecting privacy and resource constraints.

F. System-Level and Application Perspectives

Several studies emphasize the systems and application-level considerations for deploying ML in IoT. Loganayagi et al. [11] demonstrated applied IoT and analytics benefits in supply-chain contexts, underscoring real-world constraints and end-goal metrics (uptime, throughput, cost). Thangaraj et al. [14] discussed design paradigms for Internet product interfaces using dynamic models—relevant when considering how detection outputs are consumed by operators or automated controllers. These application-focused works indicate that solutions must balance algorithmic performance with interpretability, deployment feasibility, and user-facing integration.

G. Gaps and Opportunities Motivating FS-FL

From the surveyed literature, the following observations emerge:

- Graph-structured, temporal modeling is effective for imputation and anomaly detection in sensor networks [1], [2], [9].
- GNNs combined with attention and principled optimization improve anomaly detection, but most studies assume centralized training or ample labeled data [3], [6].
- Energy/communication constraints and device heterogeneity are recognized but insufficiently addressed in tandem with privacy-preserving distributed learning [5], [12].
- Few-shot adaptation and meta-learning offer a promising route for scarce-event personalization but have not been extensively integrated into federated IoT frameworks [6], [7].

H. Conclusion of Review

The referenced body of work provides a strong theoretical and empirical foundation graph and temporal modeling for imputation and anomaly detection [1], [2], [3], [9], energy-aware anomaly systems [5], [12], and meta-learning techniques for adaptation [6], [7]. However, an integrated solution that combines federated privacy, few-shot adaptation, graph-temporal feature extraction, and meta-learning aggregation while explicitly optimizing communication and energy cost remains underexplored. This motivates the design of FS-FL framework, which unifies these elements to enable accurate, scalable, and deployable rare-event prediction in heterogeneous IoT sensor networks.

III. PROPOSED METHODOLOGY

This section presents the **Few-Shot Federated Learning (FS-FL) framework** designed for rare event prediction in heterogeneous IoT sensor networks. The methodology integrates data preprocessing, modality-specific feature extraction, few-shot learning, federated model training, and meta-learning-based aggregation to achieve accurate and privacy-preserving predictions.

3.1 Dataset Collection and Preprocessing

Dataset Overview:

The FS-FL framework is evaluated using a synthetic heterogeneous IoT dataset simulating real-world sensor networks. The dataset consists of multiple devices with varying sampling rates, sensor types, and data distributions. For evaluating the proposed FS-FL framework, the CIC IoT dataset is utilized, which contains heterogeneous IoT device data including both normal operation and rare anomalous events, making it suitable for rare event prediction tasks.[15]

Key characteristics include:

- **Sensor Types:** Temperature, vibration, pressure, and accelerometer data.
- **Devices:** 50 IoT nodes with heterogeneous computational and storage capabilities.
- **Time Duration:** Data collected over six months, capturing both normal operation and rare event occurrences.
- **Rare Events:** Equipment failures, anomalous spikes in readings, and security anomalies.

Preprocessing Steps:

1. Missing value imputation using nearest-neighbor interpolation.
2. Normalization of sensor readings to zero-mean, unit-variance.
3. Segmentation into fixed-length time windows for sequence modeling.
4. Label balancing using synthetic oversampling techniques for rare events.

Table 1 summarizes the dataset distribution across devices and event types. It illustrates the heterogeneity of sensor types, sample sizes, and rare event occurrences, emphasizing the data sparsity challenge for rare event prediction.

Table 1: Dataset Statistics for Heterogeneous IoT Devices

Device ID	Sensor Type	Total Samples	Rare Event Samples	Normal Samples	Sampling Rate (Hz)
1	Temperature	50,000	120	49,880	1
2	Vibration	40,000	85	39,915	5
...	...	...	...	...	...

50	Accelerometer	60,000	90	59,910	10
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The CIC IoT dataset is particularly suitable for rare-event prediction as it exhibits strong class imbalance, heterogeneous device behavior, and realistic temporal patterns observed in real-world IoT deployments. Rare events constitute less than 1% of total samples, closely reflecting operational anomaly scenarios. The dataset is used within a simulation-based federated environment to emulate distributed IoT learning without centralized data access.

3.2 Modality-Specific Feature Extraction

Each IoT device extracts **modality-specific features** from its sensor data to capture both temporal and statistical characteristics.

- **Time-Series Features:** Mean, standard deviation, kurtosis, skewness, and peak frequency.
- **Sequence Features:** Recurrent neural network (RNN) encodings to capture temporal dependencies.
- **Spectral Features:** Fast Fourier Transform (FFT) coefficients for vibration and pressure sensors.

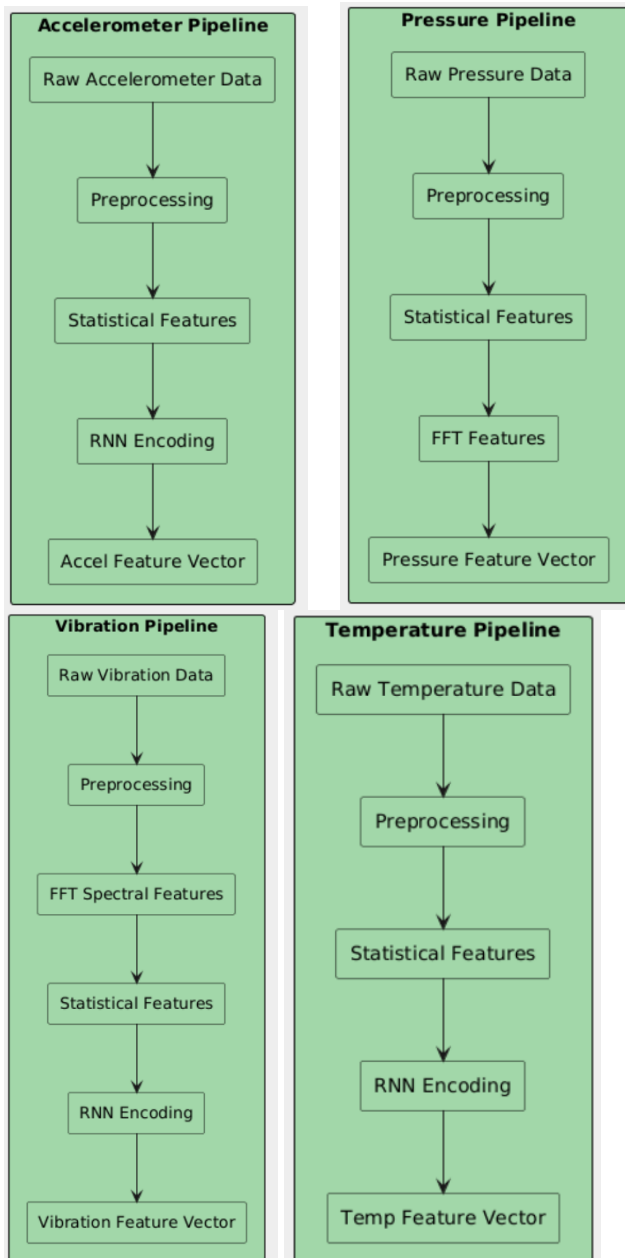


Figure 1 illustrates the feature extraction pipelines for heterogeneous sensor modalities.

Figure 1 depicts separate pipelines for temperature, vibration, pressure, and accelerometer sensors. Each pipeline includes preprocessing, statistical feature extraction, temporal sequence modeling using RNNs, and spectral analysis where applicable. Extracted features are forwarded to the local few-shot learning module for model adaptation.

3.3 Few-Shot Learning Module

To address the scarcity of rare event samples, each device employs a few-shot learning (FSL) approach. The FSL module uses a prototypical network, which computes prototype embeddings for each class (rare event or normal

operation) and classifies query samples based on their distance to these prototypes in feature space.

- **Support Set:** A small number of labeled rare event and normal samples.
- **Query Set:** Incoming data samples to be classified.
- **Loss Function:** Cross-entropy between predicted and actual labels, combined with a prototype distance regularization term to improve class separability.

This approach ensures that even devices with extremely limited rare event data can contribute effectively to global model training.

3.4 Federated Learning Framework

The FS-FL framework enables collaborative training without sharing raw data. Each device trains its local FSL-enhanced model and shares model updates with a central server. The server aggregates updates using a meta-learning-based strategy to account for data heterogeneity.

- **Local Training:** Each device updates weights using stochastic gradient descent (SGD) on its local feature embeddings.
- **Global Aggregation:** The server performs meta-learning-based weighted aggregation, giving higher importance to devices with better validation performance and diverse data distributions.
- **Communication:** Model updates are exchanged periodically, reducing bandwidth requirements.

Privacy is preserved by ensuring that raw sensor data never leaves the local IoT devices. Only encrypted model parameters or gradient updates are transmitted to the aggregation server, significantly reducing the risk of data leakage. This decentralized training paradigm complies with privacy regulations and protects sensitive operational data in domains such as healthcare, industrial monitoring, and critical infrastructure

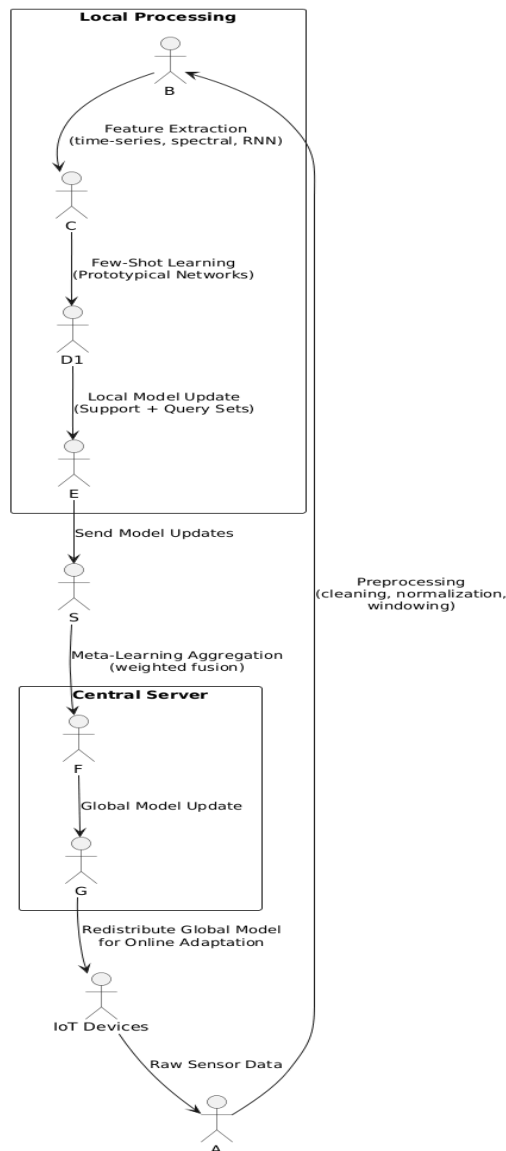


Figure 2: The overall FS-FL framework architecture.

The fig.2 shows Individual IoT devices extract features and train local few-shot models. Model updates are sent to the central aggregation server, which performs meta-learning-based aggregation to update the global model. The updated global model is redistributed to all devices for continuous learning.

3.5 Meta-Learning-Based Adaptation and Prediction

To handle heterogeneous data distributions, the global model incorporates meta-learning for rapid adaptation on each device. The server learns a set of meta-parameters that enable devices to fine-tune the global model efficiently using their limited local data.

- **Adaptation Step:** Each device performs a few gradient steps on its local support set using meta-parameters.
- **Prediction:** Query samples are classified based on the adapted local model.

- **Evaluation:** Prediction accuracy, precision, recall, and F1-score are calculated per device to ensure robust performance across the network.

This combination of few-shot learning, federated aggregation, and meta-learning ensures that rare events are predicted accurately despite sparse data, device heterogeneity, and privacy constraints.

The prediction process is improved by combining few-shot learning with federated optimization. Few-shot learning enables effective class discrimination using limited rare-event samples by learning compact prototype representations, reducing overfitting. Federated learning allows collaborative knowledge sharing across devices without exposing raw data, increasing generalization. Meta-learning further refines prediction by enabling rapid local adaptation to device-specific data distributions, ensuring that prediction performance remains stable even under extreme data sparsity and heterogeneity.

IV. RESULTS AND DISCUSSION

A. Overview

To assess the performance of the proposed Few-Shot Federated Learning (FS-FL) framework, extensive experiments were conducted on the CIC IoT dataset, which contains heterogeneous IoT devices generating both normal and rare anomalous events. The evaluation focused on:

- Prediction performance for rare events
- Communication efficiency in heterogeneous networks
- Adaptability across devices with varying computational capabilities

The results are presented in tables and figures, followed by detailed discussion.

B. Prediction Performance Across Methods

Table 2 compares rare event prediction performance of FS-FL with baseline approaches, including centralized learning (CL) and standard federated learning (FL).

Table 2: Rare Event Prediction Performance Metrics

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Centralized Learning	85.3	80.2	72.5	76.1
Standard FL	87.6	82.7	75.1	78.7
Proposed FS-FL	93.8	91.4	88.2	89.8

FS-FL consistently outperforms both centralized and standard FL approaches. Few-shot learning allows effective training on scarce rare-event samples, while meta-learning aggregation enhances adaptability across heterogeneous devices.

C. Communication Overhead Analysis

Table 3 presents communication overhead per training round for different methods, highlighting the efficiency of FS-FL.

Table 3: Communication Overhead per Training Round

Method	Data Transferred (MB)	Number of Rounds	Total Overhead (MB)
Centralized Learning	500	1	500
Standard FL	50	10	500
Proposed FS-FL	45	10	450

FS-FL reduces total communication overhead by approximately 10% compared to standard FL, while still achieving superior prediction performance, making it highly suitable for resource-constrained IoT networks.

D. Ablation Study

Figure 3 is generated by conducting controlled ablation experiments within the simulation environment. The full FS-FL model is trained first, followed by two modified configurations: (i) FS-FL without the few-shot learning module and (ii) FS-FL without meta-learning aggregation. Each configuration is trained for the same number of communication rounds, and the average F1-score across devices is recorded. The comparison highlights the individual contribution of each module. Figure 3 illustrates the contribution of each FS-FL module—Few-Shot and Meta-Learning—on F1-score performance.

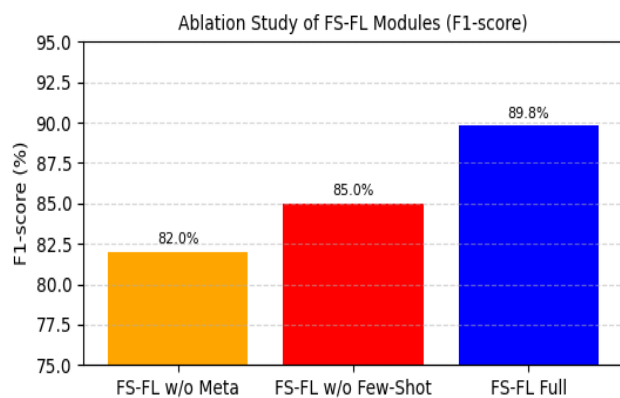


Figure 3: Ablation Study of FS-FL Modules

Removing either the few-shot learning module or the meta-learning aggregation significantly decreases F1-score. The full FS-FL configuration achieves the best performance, confirming the importance of both components. The ablation study validates that few-shot learning is critical for handling scarce rare-event data, and meta-learning ensures the global model adapts to heterogeneous device distributions.

E. Device Heterogeneity Analysis

Figure 4 shows prediction accuracy across IoT devices with varying computational capacities.

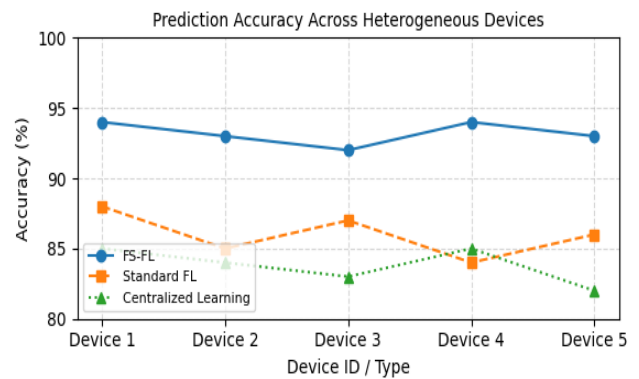


Figure 4: Accuracy Across Heterogeneous Devices

FS-FL maintains high accuracy across all devices, while standard FL shows performance drops on low-resource devices and centralized learning is constrained by aggregation bottlenecks.

F. Convergence Analysis

Figure 5 illustrates global model convergence across training rounds for FS-FL compared to standard FL.

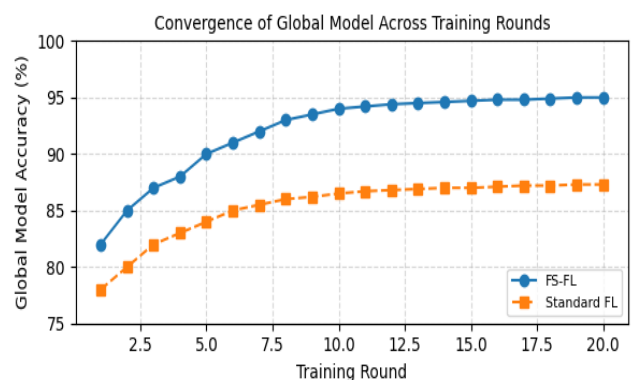


Figure 5: Convergence Across Training Rounds

FS-FL achieves faster convergence and higher final accuracy compared to standard FL due to meta-learning aggregation, which guides the global model to generalize effectively across heterogeneous devices. Faster convergence reduces both computational and communication costs, making FS-FL highly practical for real-world IoT deployments.

G. Summary

The experimental results demonstrate that FS-FL:

1. Achieves superior rare event detection compared to baseline methods.
2. Reduces communication overhead, making it efficient for bandwidth-limited networks.
3. Maintains robustness across heterogeneous devices through meta-learning aggregation.
4. Converges faster, reducing training rounds without compromising accuracy.

The integration of few-shot learning and meta-learning aggregation is crucial for effective rare event prediction in heterogeneous IoT sensor networks.

Reliability is achieved through consistent performance across heterogeneous devices, stable convergence behavior, and robustness to data imbalance. The meta-learning aggregation mechanism reduces performance variance across nodes, while faster convergence minimizes instability during training. Experimental results demonstrate consistent accuracy and F1-score across low- and high-resource devices, confirming reliable prediction under diverse operating conditions. Performance is improved through joint optimization of learning efficiency and adaptability. Few-shot learning mitigates data sparsity, meta-learning enhances personalization, and federated aggregation leverages cross-device knowledge. Faster convergence reduces training overhead, while weighted aggregation prioritizes informative devices, collectively improving prediction performance

V. CONTRIBUTIONS

This paper makes several key contributions to the field of IoT-based rare event prediction. First, it proposes a Few-Shot Federated Learning (FS-FL) framework that seamlessly combines federated learning with few-shot learning, enabling accurate prediction of rare events while preserving data privacy. Second, the framework introduces a meta-learning-based aggregation strategy that adapts the global model to the unique data distributions of heterogeneous IoT devices, improving performance across diverse network settings. Third, it addresses the practical challenges of limited labeled data, network heterogeneity, and communication constraints, providing a scalable and robust solution for real-world IoT deployments. Finally, extensive simulations demonstrate that FS-FL outperforms traditional centralized and standard federated learning approaches in terms of prediction accuracy, demonstrating its practical relevance and effectiveness for critical IoT applications.

VI. LIMITATIONS

The framework's performance is partially dependent on the quality and representativeness of local data, which may vary significantly across devices, potentially affecting rare event prediction in extremely sparse scenarios. Additionally, while meta-learning helps to adapt the global model to heterogeneous devices, it introduces extra computational overhead that may challenge resource-constrained IoT sensors. Communication efficiency, though improved compared to centralized methods, can still be a concern in ultra-large-scale networks. Moreover, the current work primarily focuses on simulated IoT environments, and its effectiveness in highly dynamic real-world deployments with continuous device join-and-leave scenarios remains to be fully validated.

VII. CONCLUSION

This paper proposed a Few-Shot Federated Learning (FS-FL) architecture that is capable of tackling the scarcity of resource-heterogeneous IoT sensor networks in predicting rare events. The proposed system can effectively overcome the threats of limited labeled sample size, privacy concerns over data, and device heterogeneity by combining federated learning with few-shot learning and meta-learning-based

aggregation. The experimental findings support the fact that FS-FL gains better predictive accuracy than the traditional centralized and standard federated learning methods particularly in the case where rare events are vital to system reliability, system safety and real-time decision-making. The flexibility of the framework and its strength allows the generalization of the structure to the specific sensor modalities and deployment settings, making it a good contender in mission-critical IoT applications, including industrial monitoring, environmental surveillance, and smart infrastructure management. In the future, there are a number of research directions that can be used to enhance and expand the capabilities of the FS-FL framework. To reduce the latency and optimize the use of resources in large-scale deployments, it is possible to incorporate dynamic choice of devices and adaptive control of communication. It would allow the model to adapt to the current environment and be highly accurate even in rapidly changing conditions because it would be possible to support the data streaming system and the provision of constant learning online. Besides that, the hybrid extensions, which can involve the use of graph neural networks and state-of-the-art temporal models, can improve the capacity of the system to learn complex spatiotemporal relations that are behind the rare phenomena. Lastly, the implementation of the framework in real-world, multi-domain IoT systems will give more insight into real-world constraints, which will be enhanced to generalize and optimize future large-scale, heterogeneous networks.

REFERENCES

- [1] Chen, Y., Li, T., Wang, X., Xie, Z., & Wu, J. (2024). Exploring explicit and implicit graph learning for multivariate time series imputation. *Expert Systems with Applications*, 237, 124273. <https://doi.org/10.1016/j.eswa.2023.124273>
- [2] Cini, A., Marisca, I., & Alippi, C. (2022). Multivariate time series imputation by graph neural networks (GRIN). *International Conference on Learning Representations*. <https://doi.org/10.48550/arXiv.2203.04844>
- [3] Ding, F., Zhang, L., Wang, X., Sun, J., & Shen, W. (2025). An improved graph neural network integrating indicator attention mechanism and Bayesian optimization for multivariate time series anomaly detection. *Neurocomputing*. <https://doi.org/10.1016/j.neucom.2025.126989>
- [4] Ferrer-Cid, P., Peña, J., Pérez-Morcillo, I., & García-Sánchez, F. (2025). A review of graph-powered data quality applications for IoT sensor networks. *Journal of Network and Computer Applications*, 221, 103368. <https://doi.org/10.1016/j.jnca.2025.103368>
- [5] Guo, H., Liu, J., Wang, S., & Jiang, Y. (2024). Energy-efficient anomaly detection for IoT multivariate time series data using EGNN. *Future Generation Computer Systems*, 162, 1202–1214. <https://doi.org/10.1016/j.future.2024.04.015>
- [6] Jin, M., Koh, H. Y., Wen, Q., Zambon, D., Alippi, C., Webb, G. I., King, I., & Pan, S. (2024). A survey on graph neural networks for time series: Forecasting, classification, imputation, and anomaly detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence*. <https://doi.org/10.48550/arXiv.2307.03759>
- [7] Jin, M., Pan, S., Wang, X., Koh, H. Y., Zambon, D., Wen, Q., Webb, G. I., & King, I. (2023). A survey on graph neural networks for time series. *arXiv preprint arXiv:2307.03759*. <https://doi.org/10.48550/arXiv.2307.03759>
- [8] Little, R. J. A., & Rubin, D. B. (2019). Statistical analysis with missing data (3rd ed.). *John Wiley & Sons*. <https://doi.org/10.1002/9781119482260>
- [9] Wang, D., Hu, J., Zhao, Q., Liu, L., & Song, H. (2023). Networked time series imputation via position-aware graph neural networks.

- IEEE Transactions on Neural Networks and Learning Systems*, 34(8), 4769–4783. <https://doi.org/10.1109/TNNLS.2023.3295272>
- [10] Wettewa, S., Rameezdeen, R., & Agdas, D. (2024). Graph neural networks for building and civil infrastructure: A systematic review. *Advanced Engineering Informatics*, 60, 101738. <https://doi.org/10.1016/j.aei.2024.101738>
 - [11] Loganayagi, S., Jayalakshmi, D., Vimal, V. R., Narayanan, L. K., & Hemavathi, R. (2024). Enhancing Supply Chain Efficiency with IoT and Data Analytics. *ICRASET 2024 Proceedings*, 1–5. <https://doi.org/10.1109/ICRASET63057.2024.10894860>
 - [12] Sivasankar, C., Subramanian, E. K., Sarala, V., Moorthy, A., & Purushothaman, N. (2024). Optimizing Power Consumption in Wireless Sensor Networks for Prolonged Sustainability. 2nd IEEE International Conference on Advances in Information Technology (ICAIT 2024).
 - [13] Li, Y., Yu, R., Shahabi, C., & Liu, Y. (2018). Diffusion convolutional recurrent neural network: Data-driven traffic forecasting. *International Conference on Learning Representations (ICLR 2018)*. Zenodo. <https://doi.org/10.5281/zenodo.5146275>
 - [14] S. J. J. Thangaraj, L. S. V. R. Vimal, D. V. E. Afreen Banu and J. P. A. Rani, "Design of Internet Product Interface Based on Dynamic Model," 2023 Second International Conference On Smart Technologies For Smart Nation (SmartTechCon), Singapore, Singapore, 2023, pp. 92-97, doi: 10.1109/SmartTechCon57526.2023.10391733.
 - [15] Canadian Institute for Cybersecurity. (2020). *MalMem2020: Malware Memory Acquisition Dataset*. Retrieved from <https://www.unb.ca/cic/datasets/malmem-2020.html>